AI Prompt Library

Copy-paste prompts for ChatGPT, Claude, Gemini or Copilot. Replace the text in [square brackets].

RULE: Use AI as a helper, not as the author or the data source. You must run the code, read the output yourself, and verify every claim and reference. Paste the 'RULES FOR YOU' line at the start of each conversation.

0. Start-of-conversation prompt (use this first, always)

```
I am a B.E. AI&ML student building the computer-vision part of an IoT agricultural rover (ESP32-CAM on a
    pan-tilt mount, tomato crop). We are writing a 6-page IEEE conference paper. I need help with
    practical steps, code and explanations.
RULES FOR YOU: Do not invent numbers, citations, library functions or results. If you are unsure, say
    so. Give short, tested steps and explain each in one line. Ask me a clarifying question if my
    information is not enough.
```

1. Hardware and firmware

1.1 Flash and test the ESP32-CAM

```
I have an AI-Thinker ESP32-CAM with an ESP32-CAM-MB USB board and Arduino IDE. Give me exact steps to
    flash the CameraWebServer example, find the IP, and test /capture. List the 5 most common errors
    (brownout, camera init failed, upload failed) and the fix for each.
```

1.2 Fix brownout / resets

```
My ESP32-CAM shows [paste Serial Monitor text]. I power it from [USB port / adapter]. Servos are powered
    [how]. Diagnose the likely causes in order of probability and give me a multimeter test for each.
```

1.3 Pan-tilt HTTP endpoint

```
Extend the Arduino ESP32 CameraWebServer example with an HTTP handler /look?pan=<0-180>&tilt=<40-140>
    that moves two hobby servos on GPIO 13 (pan) and GPIO 14 (tilt) using the ESP32Servo library,
    avoiding the PWM timer used by the camera clock. Clamp the angles. Return JSON with the angles set.
    Show only the code I must add and where to put it.
```

1.4 Wiring review

```
Here is my wiring list: [paste]. Check it for mistakes (5 V vs 3.3 V, common ground, strapping pins on
    ESP32-CAM, current needs) and tell me what could damage the board.
```

2. Data collection

2.1 Field protocol review

```
Here is my plan for collecting tomato leaf images in a farm with an ESP32-CAM and a phone: [paste
    Section 2 of file 03]. Find gaps that could cause data leakage, bias or poor labels, and suggest
    improvements I can do in 4 days.
```

2.2 Metadata sheet check

```
I will log images with these columns: [paste header of dataset_log.csv]. What important columns for a
    plant-disease research dataset am I missing? Keep it to what I can realistically record in the
    field.
```

3. Dataset preparation

3.1 Class-name mapping

```
Here are the class folder names of PlantVillage tomato: [list] and PlantDoc tomato: [list]. Propose a
    mapping table to a common class list. Mark classes that exist in only one dataset. Output CSV.
```

3.2 Leakage audit script

```
Write a Python script that reads dataset_split/train|val|test and a metadata.csv with columns file,
    plant_id and reports any plant_id in more than one split, and any duplicate file hash across splits.
    Use only the standard library plus hashlib.
```

4. Training and evaluation

4.1 Explain my results

```
These are my classifier results: [paste JSON or table]. Explain each metric in simple words, say which
    numbers look suspicious (too high / too low), and what checks I should do before trusting them. Do
    not change any number.
```

4.2 Binary healthy-vs-diseased metrics

```
Given a confusion matrix [paste] with class names [list] where 'healthy' is class [index], write a short
    Python function that computes binary accuracy, precision and recall for 'diseased' by merging all
    disease classes. I will verify it by hand.
```

4.3 Latency measurement

```
Write a Python snippet to measure median and 95th percentile inference time (ms) of an Ultralytics
    YOLOv8n model over 100 runs after warm-up on [CPU / Colab T4], using time.perf_counter. Also tell me
    what to write in the paper about the hardware.
```

4.4 Debugging training

```
My training shows [paste log]. Training accuracy is [x] but validation accuracy is [y]. List the most
    likely causes (overfitting, leakage, class imbalance, wrong preprocessing) and one check for each.
```

4.5 Cross-domain gap analysis

```
My model gets [a]% on PlantVillage-style test images but [b]% on PlantDoc field images and [c]% on my
    ESP32-CAM images. Suggest reasons for the gap and 3 cheap experiments I can run in two days to test
    each reason.
```

5. Figures and tables

5.1 Publication-quality chart

```
Write matplotlib code that draws a grouped bar chart with error bars for accuracy (mean +/- std over 3
    seeds) on three test sets: [values]. Single IEEE column width (8.8 cm), font size 8, colour-blind-
    friendly colours, axis labels with units, saves a 300 dpi PNG. Use only my numbers.
```

5.2 Confusion matrix plot

```
Write matplotlib code to plot a normalised confusion matrix from this list-of-lists [paste] with class
    names [list], readable at single-column width, annotated with percentages, saved at 300 dpi.
```

5.3 LaTeX/Word table

```
Turn this CSV into a compact IEEE-style table I can paste into Word: [paste]. Keep my numbers exactly.
    Use 2-3 decimal places consistently.
```

6. Writing support (you must supply the real facts)

6.1 Turn my notes into a method paragraph

```
Rewrite the following notes into one clear IEEE-style paragraph (about 120 words, past tense, third
    person, no first-person 'we did it for you' claims). Use ONLY facts in my notes. Mark anything
    unclear with [CHECK]. Notes: [paste your real notes]
```

6.2 Honest limitations

```
Based on these results and problems [paste], draft 4 limitation sentences a reviewer would find fair and
    honest. Do not soften or hide anything.
```

6.3 Reviewer simulation

```
Act as a strict reviewer for an IEEE conference on sustainable computing. Here is my results section:
    [paste]. List the 8 hardest questions or objections and what extra evidence would answer each.
```

7. Literature (verify everything yourself)

7.1 Search plan

```
Give me 15 search queries and 5 databases to find recent (2021-2026) papers on: tomato leaf disease
    classification with lightweight CNNs, plant disease detection domain shift to field images, low-cost
    agricultural rovers with cameras, and IoT links for farms. Do NOT list paper titles; I will find and
    verify them myself.
```

7.2 Summarise a paper I opened

```
I read this paper: [paste abstract and your notes]. Summarise in 3 sentences: method, dataset, main
    result. Then give one sentence on how it differs from a low-cost rover with two-stage vision. Use
    only the text I pasted.
```

7.3 Reference formatting

```
Format this reference in IEEE style exactly: [paste details copied from the publisher page]. Do not add
    or change any facts.
```

8. Quick sanity prompts

8.1 Is this claim safe to write?

```
Here is a sentence I want to write in the paper: [sentence]. Here is the evidence I have: [files /
    numbers]. Is the claim fully supported? If not, rewrite it so it only claims what the evidence
    shows.
```